

PROJECT OVERVIEW

This project presents the design and development of a Communicable Fault Passage Indicator (FPI) for low-voltage (LV) distribution networks in Bangladesh. The system detects fault currents, identifies fault passage, and communicates real-time alerts to a central server for faster fault isolation and restoration. The device is solar rechargeable, weatherproof, easy to install, and cost-effective, aiming to reduce outage duration, technical losses, and improve grid reliability.

PROBLEM STATEMENT

Bangladesh's largely unmonitored LV distribution network, affected by aging conductors, overloading, illegal connections, and environmental faults, experiences high technical (10–15%) and non-technical (5–10%) losses with 3–4 hour restoration times causing outages, higher SAIDI (12–15 hours/year), ENS and economic penalties which can be reduced through smart real-time monitoring, automated fault detection, and load forecasting.

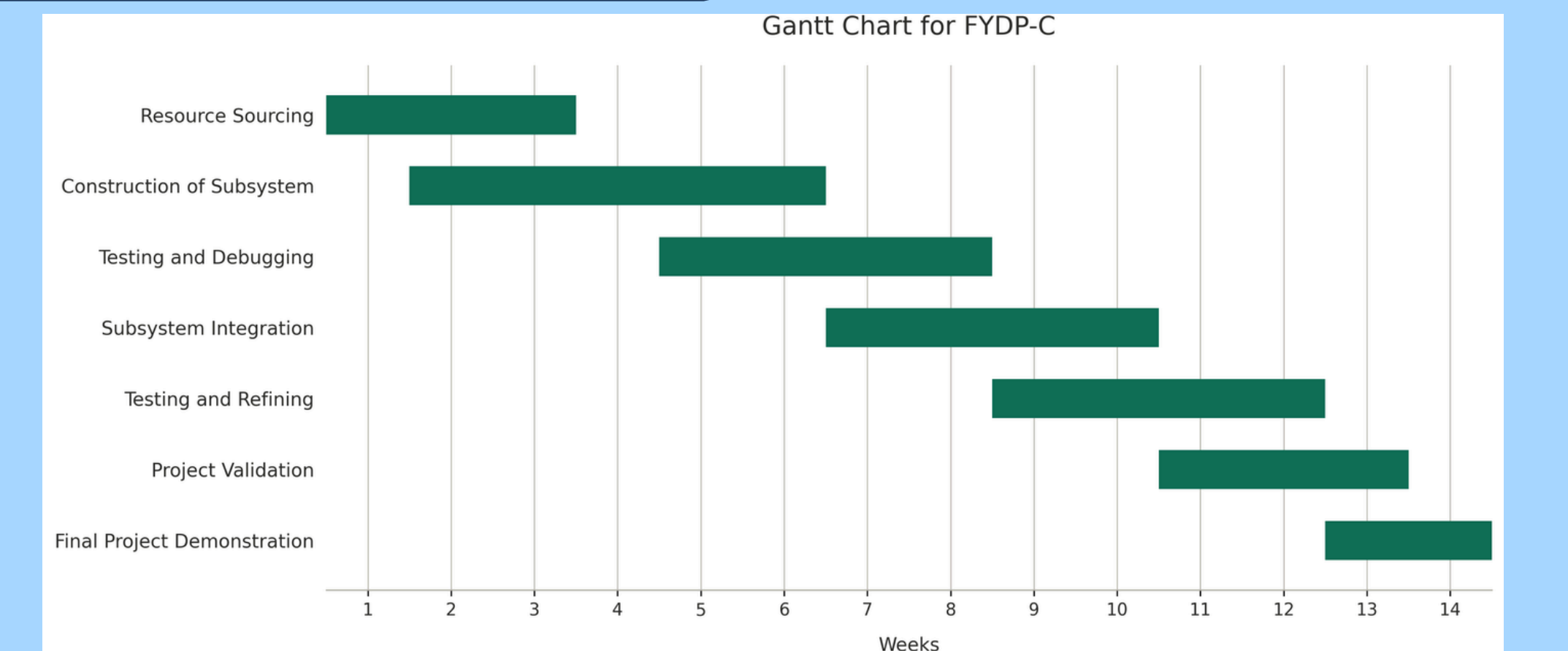
OBJECTIVE

- Design of an intelligent Fault Passage Indicator (FPI) system with adaptive algorithms for fault detection and identification.
- Enable communication with a central server for data transmission.
- Validate system performances on fault conditions and analyze data.

SYSTEM REQUIREMENTS

Requirements	Constraints	Specifications
Real-time current monitoring for fault detection	LV current fluctuations harder to analyse than MV faults	Voltage Level: 12–24 V AC
Fast response time (0.2–0.5 s)	Low latency requirement (<150 ms)	Fault Current Range: 0–20 A
Detect and classify fault type	Fault must sustain for 40–100 ms	Fault Detection Time: Less than 5 ms
Distinguish fault-induced current from normal load variation	Dependent on communication infrastructure (LoRaWAN/LTE availability)	Indication Time: 1–2 sec
Electricity theft / current comparison between poles	EMI must be less than 10 V/m around the FPI	Battery Life: Up to 3 Years
Local indication (LED/buzzer)	Must meet safety standards near exposed LV lines	
Communicate fault alerts to central server	Limited budget for industrial-grade sensors/components	

PROJECT SCHEDULE



References: 1) S. H. Asman, N. F. A. Aziz, U. A. U. Amirulddin, and M. Z. A. Kadir, "Transient Fault Detection and location in power Distribution Network: A review of current practices and challenges in Malaysia," Energies, vol. 14, no. 11, p. 2988, May 2021, doi: 10.3390/en14112988. 2) A. Farughian, L. Kumpulainen, K. Kauhaniemi, S. Pettissalo, and V. Sallinen, "Technical requirements for practical implementation of fault passage indication," in Proc. IEEE PES Innovative Smart Grid Technologies Europe (ISGT Europe), 2021, pp. 1–6, doi: 10.1109/ISGTEurope52324.2021.9640

METHODOLOGY & SYSTEM ARCHITECTURE

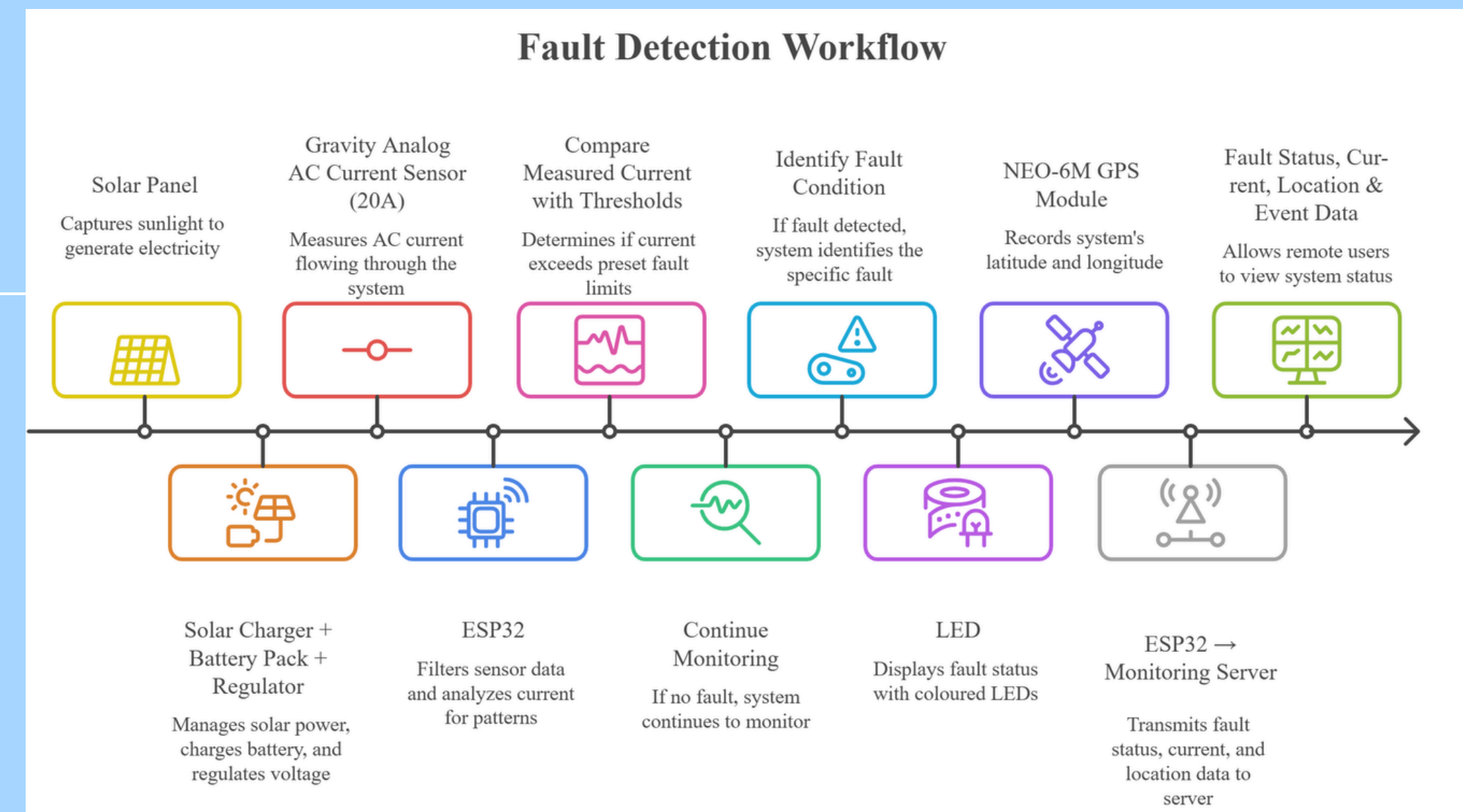


Fig 1. FPI Fault Detection Block Diagram

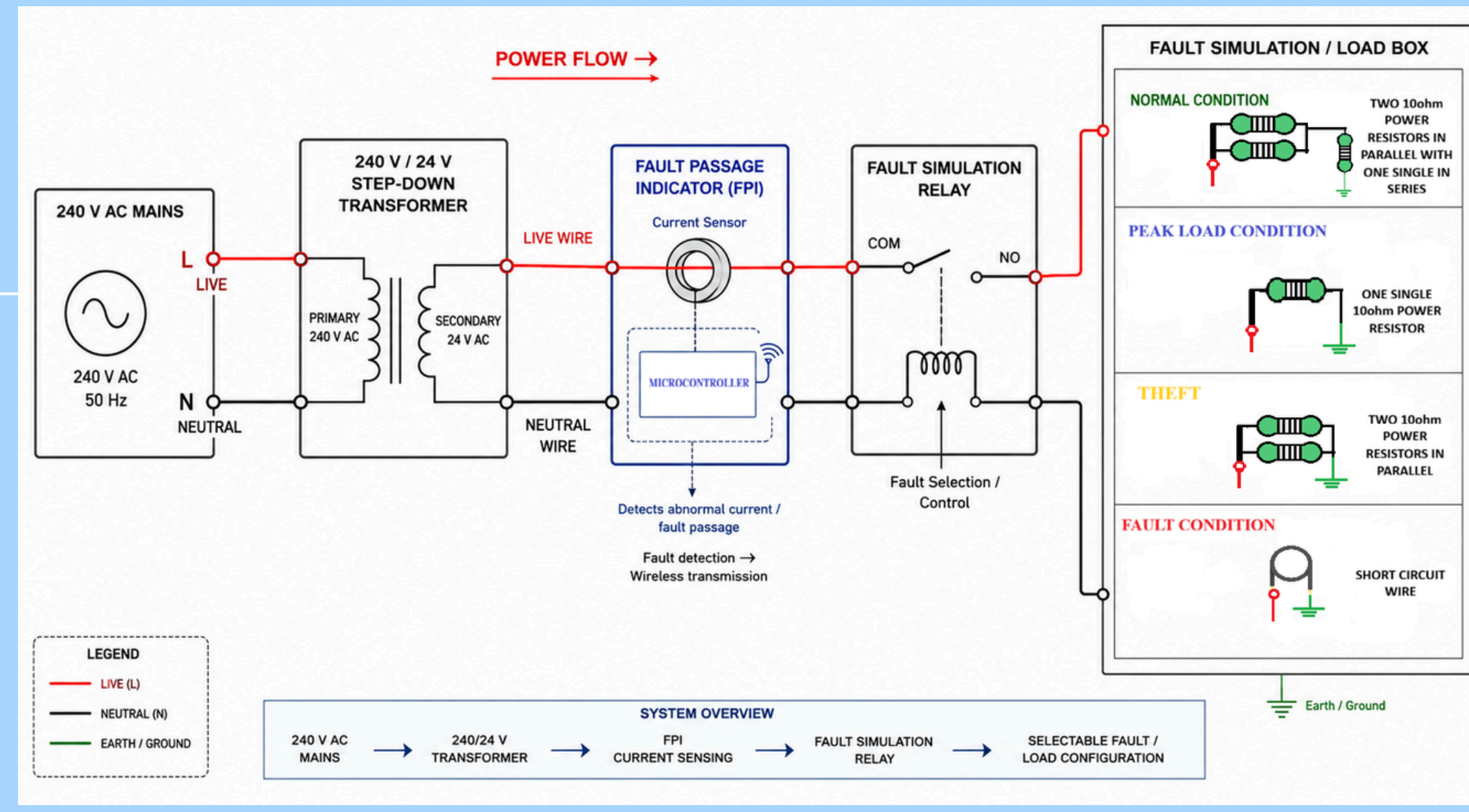


Fig 2. Overall System Design Architecture

FAULT PASSAGE INDICATOR DEVICE

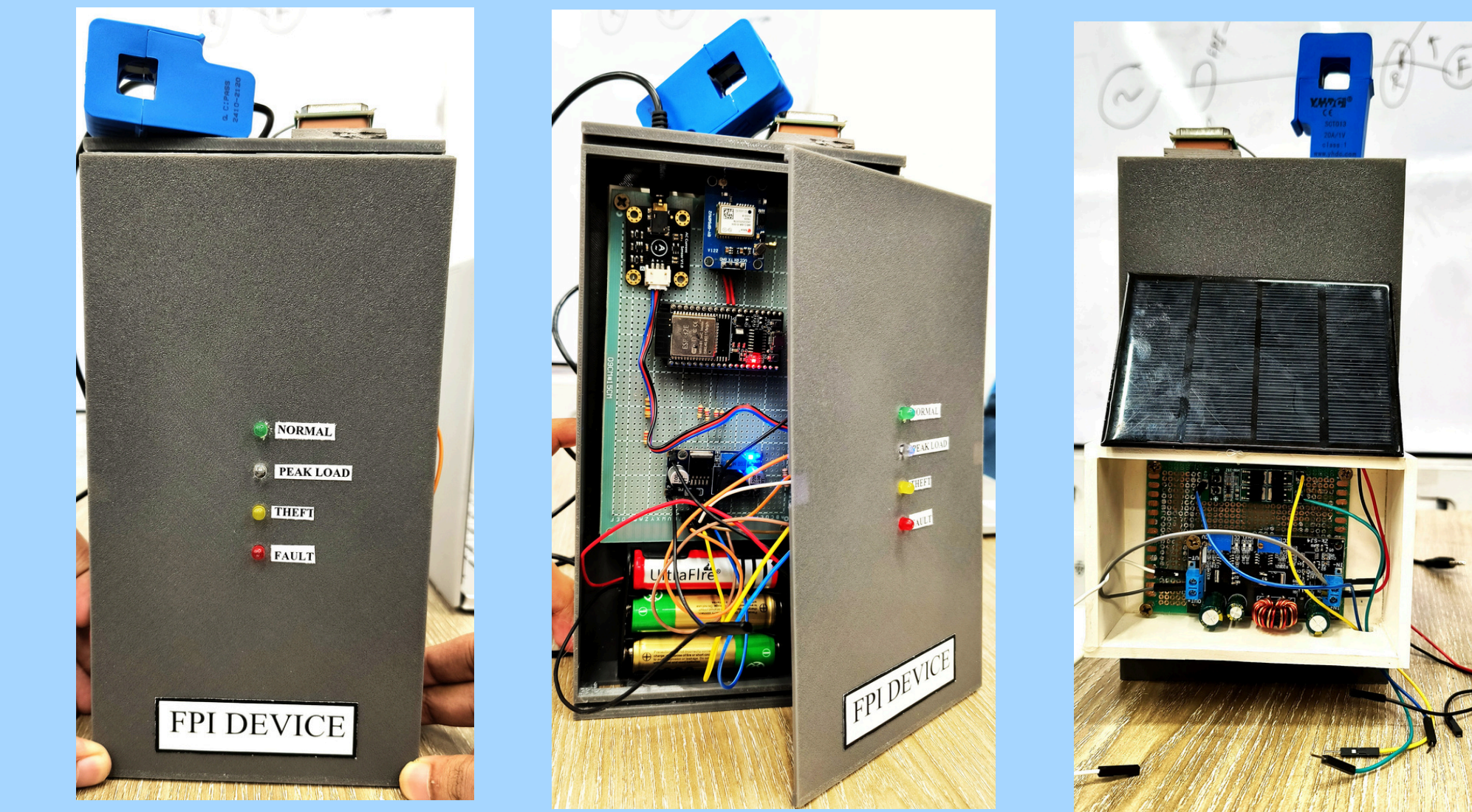


Fig.3 a) Front View

Fig.3 b) Internal Circuit View

Fig.3 c) Back View

ETAP SIMULATION

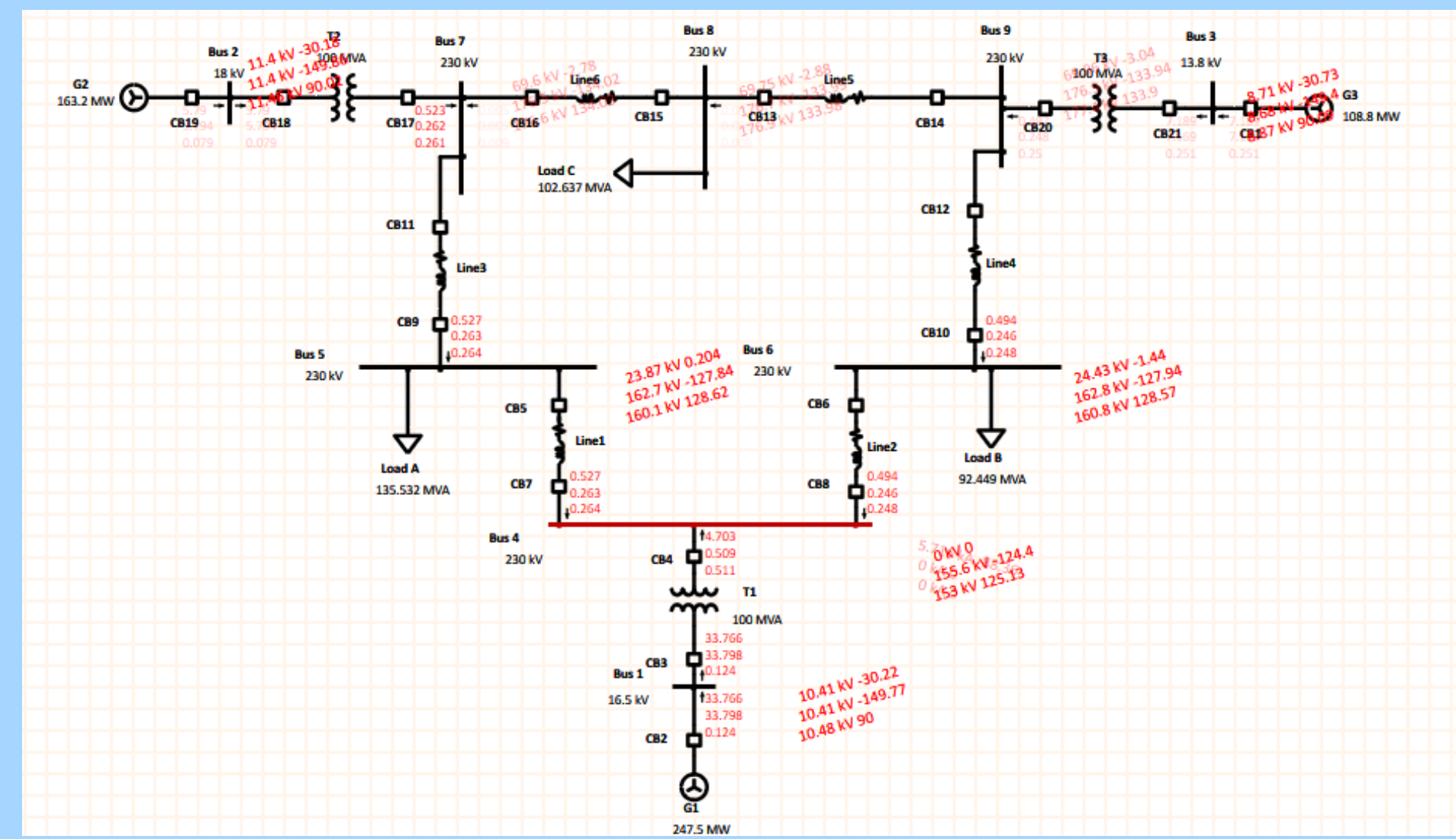


Fig 4. Fault Current Values in ETAP Simulation

WEB DASHBOARD



Fig 5. a) UI Interface

b) Current vs Time Graph

c) Current Fluctuation Graph

d) Gps Coordinates in UI

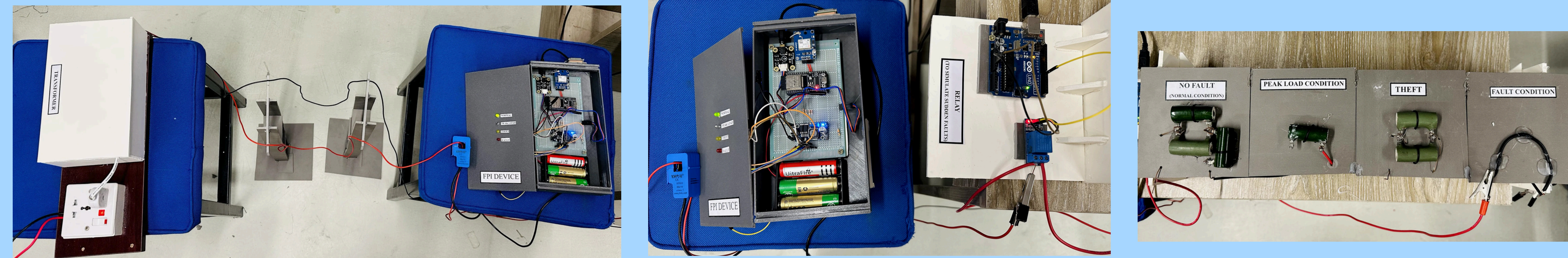


Fig 6. System Design Architecture

FUTURE WORK

- Test the system on real AC distribution networks with higher-rated sensors, and run long-term trials on actual feeders.
- Add LoRaWAN/NB-IoT/LTE-M for wider-area communication and integrate with SCADA/DMS for centralized monitoring.
- Improve fault detection with AI/ML and hybrid models, and harden the hardware for long-term outdoor/industrial use.

CONCLUSION & ACKNOWLEDGEMENT

This project presents a smart and cost-effective communicable FPI for Bangladesh's low-voltage network, enabling fast fault detection, local alerts, and remote monitoring to improve reliability and reduce outage time. We express our sincere gratitude to our supervisor and the Department of EEE, BRAC University, for their guidance and support. Special thanks to DPDC officials and field engineers for their valuable insights.

PROGRESS FROM FYDP P → D → C

Phase	FYDP P (Planning)	FYDP D (Design & Development)	FYDP C (Hybrid Model Development)
Objectives	- Literature review & requirement analysis - Define system architecture - Select components & communication tech - Plan prototype & set targets	- Simulate 2 approaches: Hall-effect + battery + Arduino + LoRa vs Rogowski coil + solar-hybrid + cloud/GSM - Compare subsystems (power, sensing, processing, comms) - Decide: hybrid of both beats either alone	- Implement hybrid: Rogowski coil + solar-hybrid power (Approach 2) + Arduino + LoRa (Approach 1) - Remove each approach's weak points (battery life, cloud latency) - Validate & prepare for deployment
Key Focus	Feasibility study	Simulation & comparison	Hybrid integration & validation
Approach	Conceptual design & planning	Two designs modelled & compared subsystem-by-subsystem	Best subsystems from both combined into one hybrid system
Outcome	Feasible plan & target specs	Decision to build the hybrid design	Industry-ready hybrid FPI system

ECONOMIC ANALYSIS & COST-BENEFIT ESTIMATION

ECONOMIC IMPACT OF COMMUNICABLE FPI IN BANGLADESH			
INDICATOR	EXISTING NETWORK	WITH FPI / POTENTIAL IMPACT	ACTUAL BENEFIT GAINED
Fault Identification	~180 min	<30 min	Power restored to customers roughly 6x faster, cutting outage duration significantly
SAIDI	12-15 h/year	60-70% potential reduction	Fewer customer-minutes lost per year, directly improving reliability scores
Major-event contribution	4-6 h	Reduced through targeted restoration	Crews go straight to the fault instead of patrolling lines, speeding up storm recovery
FPI unit cost	BDT 100k+ (imported)	BDT 20k-40k (60-80% below imports)	Lower unit cost makes wider, more affordable deployment across the grid possible
Electricity theft	~15% unaccounted	Abnormal-current monitoring	Illegal tapping is flagged in near real time instead of discovered later
Theft-related economic loss	~US\$600M/year	Potential loss reduction	Earlier theft detection translates into direct revenue recovery for utilities
Distribution losses (technical)	10-15%	Better monitoring & fault management	Proactive maintenance driven by live data lowers technical line losses
Non-technical losses	5-10%	Theft/abnormal-current detection	Real-time alerts reduce revenue leakage from non-technical losses
11 kV network length	>62,000 km	Improved monitoring coverage	Visibility extends across a much larger share of the feeder network
Deployment potential	-	500-1,000 FPIs / 50 feeders	A concrete, scalable rollout path from pilot feeders to broader coverage
SLG fault share	~70% of incidents	Better detection & faster isolation	The most common fault type gets isolated faster, limiting cascading outages

Table 3. Economic Impact Analysis

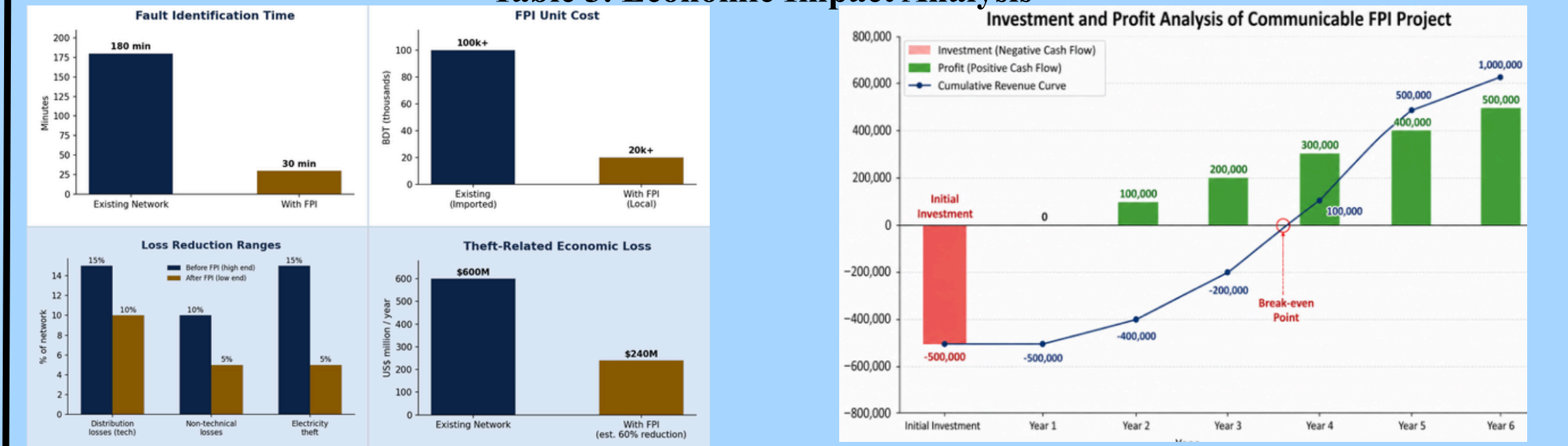


Fig. 7 Economic Impact Graphs

Subsystems	Component Name	Unit Price (BDT)	Quantity	Total Price (BDT)
Power	12V 1.5W Solar panel	150	1	150
	24V 3A Step-down Transformer	500	2	1000
	Solar charge controller - 25 BMS	920	1	920
	3.7V Rechargeable Battery	50	6	300
	DC-DC Buck Converter	250	1	250
Sensor	Gravity Analog AC Current Sensor	3500	1	3500
	Arduino ESP32	450	1	450
Indicator	100 W 10 ohm Power Resistor	300	6	1800
	LED	10	10	100
Communication	Neo-6M GPS	440	1	440
	3D Printing Filament	2700	1	2700
Miscellaneous	PCB + Wiring + Connectors + Others	1400	1	1400
Total Price				13010 BDT

Table 4. Bill of Materials

ETHICAL CONSIDERATION

- System Integrity & Compliance:** Ensures accurate detection, secure data handling, and standards compliance (IEEE C37.113, 519, 1686, 802.1X, 1547) for reliable and accountable operation.
- Safety & Ethical Responsibility:** Enables safer remote monitoring, reduces worker risk (IEEE 1584), and promotes ethical engineering through quality design, transparency, and public benefit.

